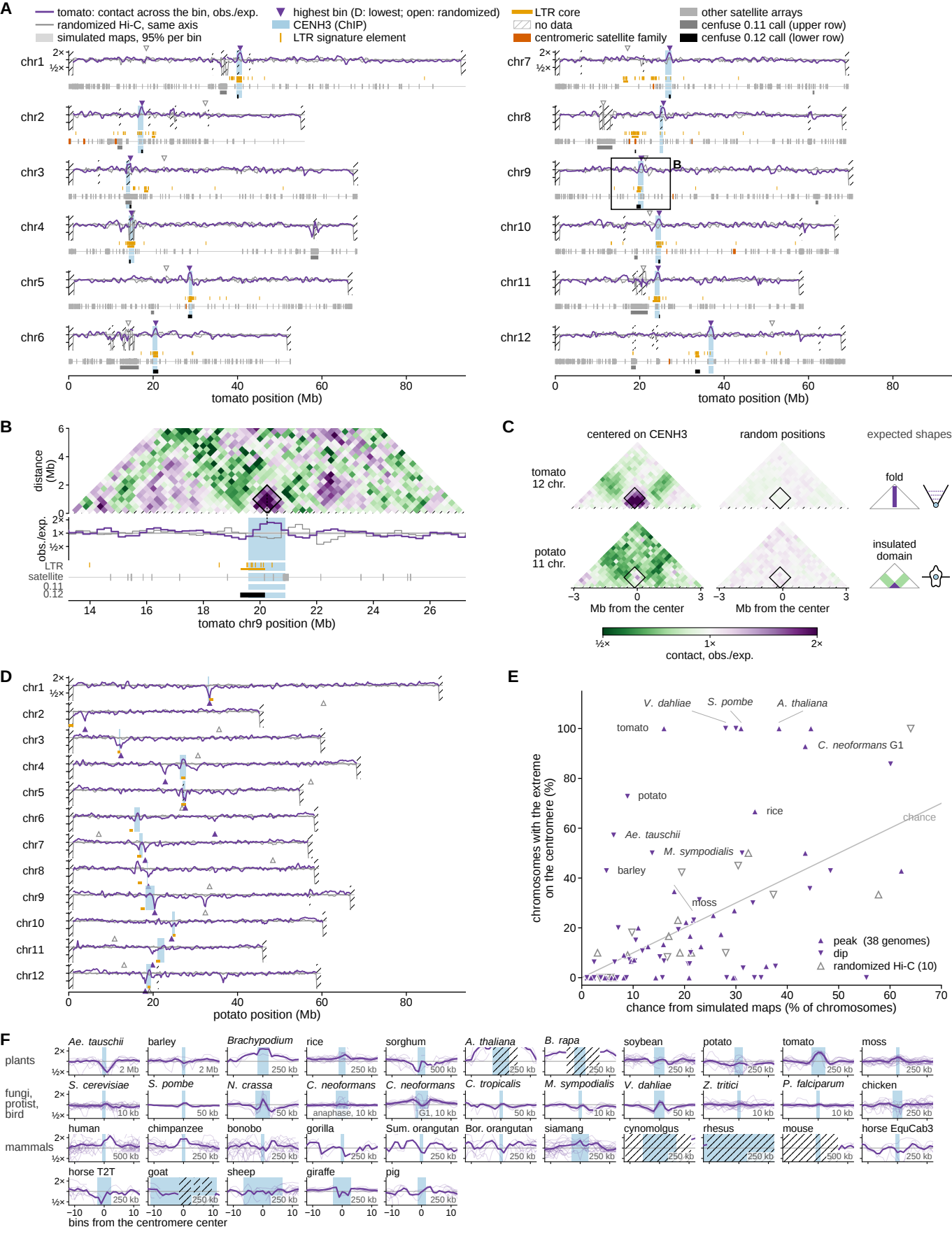

A
tomato: contact across the bin, obs./exp.
randomized Hi-C, same axis
simulated maps, 95% per bin
highest bin (D: lowest; open: randomized)
CENH3 (ChIP)
LTR signature element
LTR core
no data
centromeric satellite family
other satellite arrays
cenfuse 0.11 call (upper row)
cenfuse 0.12 call (lower row)
chr1
chr2
chr3
chr4
chr5
chr6
chr7
chr8
chr9
chr10
chr11
chr12
tomato position (Mb)
B
distance (Mb)
obs./exp.
LTR
satellite
0.11
0.12
tomato chr9 position (Mb)
C
centered on CENH3
random positions
expected shapes
tomato 12 chr.
potato 11 chr.
fold
insulated domain
Mb from the center
contact, obs./exp.
D
potato position (Mb)
E
V. dahliae
S. pombe
A. thaliana
tomato
C. neoformans G1
potato
rice
chance
Ae. tauschii
M. sympodialis
barley
moss
peak (38 genomes)
dip
randomized Hi-C (10)
chromosomes with the extreme on the centromere (%)
chance from simulated maps (% of chromosomes)
F
plants
fungi, protist, bird
mammals
Ae. tauschii
barley
Brachypodium
rice
sorghum
A. thaliana
B. rapa
soybean
potato
tomato
moss
S. cerevisiae
S. pombe
N. crassa
C. neoformans
C. neoformans
C. tropicalis
M. sympodialis
V. dahliae
Z. tritici
P. falciparum
chicken
human
chimpanzee
bonobo
gorilla
Sum. orangutan
Bor. orangutan
siamang
cynomolgus
rhesus
mouse
horse EquCab3
horse T2T
goat
sheep
giraffe
pig
bins from the centromere center

The two sides of every tomato centromere contact each other about twice as often as equally distant DNA elsewhere, so a contact map alone marks centromeres that no repeat marks. Potato shows a deficit instead, and the signal recurs in plants and fungi but not in mammals.

A On all 12 tomato chromosomes, contact across the chromosome peaks once, inside the CENH3 domain; randomized Hi-C and simulated maps show no such peak. Satellite arrays and the centromere-marking retrotransposons often lie elsewhere, which is why calls made from repeats missed.

B Tomato chr9 as a contact map: the excess is a compact patch of contact between the two sides of the centromere, directly above it.

C Averaged over chromosomes, tomato centromeres gain contact across the centromere and lose contact with the rest of the chromosome, more like a small insulated domain than arms folded together; potato centromeres lose contact instead. Random positions stay flat.

D In potato the same measure dips at most centromeres instead of peaking.

E In 11 of 38 genomes with Hi-C and a centromere map, the peak or the dip falls on the centromere more often than chance: plants and fungi, never mammals.

F The average profile around the centromeres of each genome shows which genomes peak, which dip and which are flat.

Terms. Contact across a bin (purple): balanced Hi-C contact between the 4 bins to its left and the 4 bins to its right, over what the chromosome's mean contact at those distances predicts (obs./exp.; bins of 250 kb in tomato and potato). Peak, dip: its highest and lowest bin on a chromosome. On the centromere: within 4 bins of the CENH3 or CENP-A domain mapped by ChIP. Chance: how often the extreme falls there in 100 maps simulated per chromosome from its coverage and distance decay, with no special locus. Randomized Hi-C: the same read pairs with second ends permuted among pairs (coverage kept, no 3D structure). P: Poisson-binomial tail over chromosomes.

A Tomato (*Solanum lycopersicum*, Heinz 1706), all 12 chromosomes to one scale. The peak lies on CENH3 on 12/12 chromosomes (chance 0.16 per chromosome, $P = 4.6 \times 10^{-12}$); peaks reach 1.89× (median; 1.66–2.07×), simulated profiles 1.07× (gray band: 95% per bin). Randomized Hi-C (gray line, same axis) peaks on CENH3 on 2/12. Beneath: every element of the chromosome's LTR signature families (ticks) and their core (bar); every satellite array (vermillion: the family classed centromeric, on 7/12); cenfuse calls. Cores lie within 1 Mb of CENH3 on 7/12. Tested against the cores, without truth, the peak lies at the core on 7/12 (P = 0.0001), so cenfuse 0.12 uses it: 10/12 right, none wrong with high or medium confidence, against 1/12 and 7 in 0.11, whose calls followed satellites (0.12 with randomized Hi-C: 6/12, 2).

B Tomato chr9, 13–27 Mb (box in A). Top: obs./exp. contact map, each pair of bins drawn above its midpoint, up to 6 Mb apart (hatched: masked bins). The diamond is the crossing box of the peak bin: the 16 pairs whose sum, over its expectation, is the profile value beneath (1.74×), on the same axis. The excess sits in the lowest pairs of the box, those closest across the centromere.

C Obs./exp. maps averaged over ±3 Mb windows centered on CENH3 (12 tomato, 11 potato chromosomes) or on 5 random positions per chromosome at least 6 Mb from it. Tomato: 1.79× in the crossing box at the center (random 0.97×) but 1.30× for pairs on one side of it, so contact is raised within about 1 Mb and most across the centromere; the centromeric bins are depleted of contact beyond about 1 Mb (stripes). Potato: 0.92× (random 1.04×). Cartoons: a fold pairs the arms, an excess running straight up from the centromere; an insulated domain gives a small triangle on the diagonal with depleted stripes. At 250 kb the two differ mainly in how far the excess extends. The mechanism is a hypothesis; precedent: heterochromatin-dependent contacts between the arms beside fission yeast centromeres (Mizuguchi et al. 2014, Nature 516:432).

D Potato (*Solanum tuberosum*), drawn as in A (orange bar: LTR core; triangle: lowest bin). The dip lies on CENH3 on 8/11 chromosomes with a CENH3 map (chance 0.09, $P = 2.9 \times 10^{-7}$; median depth 0.40×), the peak on 1/11; chr2 has no map. Against the LTR cores the dip lies at the core on 7/12 chromosomes (P = 1.4e-05). Randomized potato Hi-C: dip on CENH3 on 2/11.

E One mark per genome and sign (38 genomes with Hi-C and a centromere map); the line is chance. Named: $P \leq 0.05$ (11 genomes: 7/11 plants, 4/9 fungi, 0/16 mammals, not the bird or *Plasmodium*). Open marks: randomized Hi-C of 10 of them, each with its own simulated chance; beyond chance in 1 (moss, dip on 11/26, P = 0.005; with 20 tests, one P this small is expected with probability 0.10). Every randomized partner fails the cenfuse test against the LTR cores; the closest is moss (P = 0.011 against 0.01). P values are not corrected for trying two signs.

F Each genome's profile around the center of its centromere maps, in its own bins (size at the lower right): one faint line per chromosome and their mean; blue: median centromere width; hatched: no mean (the centromeric bins are unmappable). Lines leaving the axis are cut.

Exceptions and choices. In soybean, sorghum and every mammal the extreme does not fall on the centromere beyond chance, although the mean profile rises at the centromeres of human, chimpanzee, siamang (F). The sign differs between genomes (peak in tomato, rice, *A. thaliana* and moss; dip in potato, barley and *Ae. tauschii*), so cenfuse picks it per genome from the LTR cores and uses the signal only when that test passes. At low coverage a profile is noisy and its extremes fall where coverage is lowest, often at centromeres, which the simulated chance accounts for. The two *C. neoformans* maps and the two horse assemblies are not independent. Example in B: among the 7 tomato chromosomes whose peak lies on CENH3 and at the LTR core, the one nearest the median of peak height and CENH3 width (each scaled by its median absolute deviation) whose map is whole around the peak (no masked bin or assembly gap within 2 Mb); chr4, nearer the medians, has masked bins at the peak. It is not the strongest, and its CENH3 domain is the ChIP domain that scores the call.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11

Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/figures/fig_cis_centromere/fig_cis_centromere.py

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